

Bipin Kumar Chaudhary

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Portfolio | LinkedIn | GitHub | LeetCode

SUMMARY

Full Stack Engineer with **3+ years** designing and shipping production systems end-to-end from serverless video pipelines on AWS to AI-enriched workflows and real-time scheduling engines. Delivered **40% latency reduction** and **30% faster queries** in live systems. M.Tech (CSE) from MNNIT Allahabad, **350+ LeetCode**, and merged open-source contributor.

EXPERIENCE

Independent Full-Stack Developer (Self-Employed) | Remote | Oct 2023 - Present

- Built and deployed production-grade applications including a video streaming platform and a clinic management system with real-world users.
- Designed end-to-end system architecture covering authentication, scheduling engines, caching strategies, and asynchronous workflows.
- Integrated third-party services (WhatsApp API, OpenAI) to enable real-time communication and AI-powered features.

SAP Labs | Graduate Research Engineer | *iXp Internship, M.Tech MNNIT Allahabad* | Bangalore Aug 2024 - Jun 2025

- Performed functional and regression testing for SAP Business Application Studio (BAS), identifying **60+ bugs** across development workflows and platform features, and documented detailed defect reports and test observations to accelerate triage and resolution.
- Developed **30+ automated test scripts** and **50+ BAS applications** to simulate diverse testing scenarios, converting manual test cases into repeatable validation pipelines with structured test case documentation for long-term maintainability.
- Validated database artifacts and HANA Vector Engine features using SAP HANA Database Explorer, ensuring correctness across **100+ test scenarios** with zero critical failures in production sign-off.

Nagarro | Full Stack Developer | *Gurgaon* May 2021 - Sep 2023

- Improved frontend performance by **20% TBT reduction** using Angular OnPush strategy and reusable component architecture, maintaining **60 FPS** under high data load.
- Designed and developed scalable backend services for a rail asset management system managing **170K+ assets**, enabling sub-second real-time updates; authored comprehensive API documentation and feature specs adopted as the team reference standard.
- Enhanced engineering productivity by standardizing CI/CD pipelines and unit testing (Jasmine), improving deployment speed by **15%**, and maintained living test documentation to onboard junior developers faster.

PROJECTS

StreamSphere | *Angular, Node.js, AWS, MongoDB, Redis* [Live] | [GitHub]

- Architected a **serverless video processing pipeline** (S3 → Lambda → FFmpeg → HLS) generating adaptive bitrate streams (360p/720p/1080p), reducing playback startup time by **40%**; documented full system architecture including data flow diagrams and component interaction specs.
- Implemented **event-driven caching** with Redis and Lambda webhooks, reducing feed latency by **30-40%** and S3 origin load by **25%**.
- Designed **cursor-based pagination** for infinite scroll, ensuring O(1) query performance and consistency under concurrent uploads.
- Built a fault-tolerant **AI enrichment pipeline** (Whisper, GPT, HuggingFace) using async orchestration, allowing graceful degradation without blocking core workflows.

DentalConnect | *Next.js, PostgreSQL (Supabase), Vercel, WhatsApp API* [Live] | [GitHub]

- Designed end-to-end **system architecture** for a full-stack appointment management platform covering auth flows, scheduling logic, notification pipelines, and observability with OTP-based WhatsApp authentication, JWT role-based access (doctor/patient), and a **dynamic scheduling engine** reconciling weekly schedules, blocked dates, partial slot blocks, and live bookings in real time.
- Engineered **asynchronous notification workflows** (OTP, confirmations, reminders, cancellations) via WhatsApp Business API and **Vercel Cron**, with strong data security using Row-Level Security (RLS), Zod validation, OTP rate limiting, and middleware-based route protection.
- Developed a **doctor dashboard** with live queue management, appointment lifecycle tracking, review moderation, and operational analytics, alongside **API observability infrastructure** for request logging, latency tracking, severity classification, and automated log retention via cron jobs.

OPEN SOURCE CONTRIBUTIONS

- GSoC Org Finder | *JavaScript, HTML/CSS* [PR #128]
- Replaced hardcoded static milestone HTML with a JS-driven rendering engine introduced a MILESTONES data layer, date-aware state classification, and live 60s re-render cycle; merged after multi-round review with edge case handling for invalid dates, UTC normalization, and DOM guard checks.
- Online Clipboard | *Next.js, React, Framer Motion* [PR #11]
- Added animated skeleton loaders for Dashboard and Community pages, replacing plain text loading states with layout-matched shimmer rows and a smooth Framer Motion fade-in on data arrival improving perceived performance across both desktop and mobile views.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, C++, Java, SQL
Core Concepts: Data Structures & Algorithms, System Design, OOP, Concurrency, Caching, Distributed System, REST APIs
Frontend: Angular, React, Next.js, HTML5/CSS3, Tailwind CSS
Backend: Node.js, Express, REST APIs, CI/CD, Docker, Git
AI/ML: OpenAI APIs, Whisper (ASR), HuggingFace Transformers, RAG, Vector Embeddings
Cloud & Databases: AWS (S3, Lambda, CloudFront), SAP BTP, SAP HANA, MongoDB, Redis, PostgreSQL

EDUCATION

- Motilal Nehru National Institute of Technology Allahabad | M.Tech(CSE) | **CPI: 8.75/10**
Includes SAP Labs iXp internship (Aug 2024 - Jun 2025) Aug 2023 - Aug 2025
- Bundelkhand Institute of Engineering and Technology | B.Tech(CSE) | **78.6%** Aug 2017 - Aug 2021